

**DATA VISUALIZATION ASSOCIATE
EARLY REMOTE INTERNSHIP**

Week 3: Dashboard and Analytical Report

Dashboard Design, SQL Validation, Key Findings, and Recommendations

Prepared by

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1. Executive Summary

Week 1 set up the database and explored the raw data. Week 2 cleaned the data and joined the two datasets together into one integrated dataset. Week 3 uses that clean, joined data to build a dashboard. This report documents the Week 3 dashboard development phase, built on the cleaned and integrated dataset (integrated_dataset) produced during Week 2. The objective was to design a Looker Studio dashboard that communicates opportunity sign-up performance, completion rates, reward distribution, application trends, scholarship distribution, and lead source performance, and to validate every metric shown on the dashboard with a corresponding SQL query.

Eight visualizations were built and independently validated against SQL query results before being added to the dashboard, ensuring every number a stakeholder sees can be traced back to a specific, reproducible query. An Eighth analysis, the percentage change in applications between Q4 2025 and Q1 2026, was also computed and validated. The findings point to a clear best-performing category by completion rate (Events), a dominant lead-acquisition channel (Social Media), and a notable recent decline in application volume worth flagging for stakeholder attention.

1.1 Objectives

- Design a Looker Studio dashboard covering sign-up performance, completion rates, reward distribution, application trends, scholarship distribution, and lead source performance.
- Build meaningful, well-labelled visualizations that make patterns and comparisons easy to read.
- Validate every key metric shown on the dashboard with an independent SQL query.
- Generate insights and recommendations grounded in the validated data, not assumption.

1.2 Data Source

This report uses the integrated_dataset table built in Week 2. It has 14,825 applications, each one linked to the specific opportunity it was submitted to. This table was already cleaned: duplicates and corrupted rows were removed, missing values were handled, category and fee were backfilled from the join which were documented in the Week 2 report.

2. Dashboard Design

2.1 Dashboard Link

[Live Looker Studio dashboard link](#)

2.2 What the Dashboard Needed to Show

The brief asked for six things. Each one became a section of the dashboard.

- Opportunity sign-up performance
- Completion rates across opportunities
- Reward distribution
- Application and sign-up trends over time
- Scholarship distribution by category
- Lead source performance

2.3 Layout

The dashboard is in pages, with each page laid out in two rows. The top row has Scorecards for a quick summary: total applications, total opportunities, overall completion rate, and total category. followed by the eight required visualizations which are bar charts comparing categories and opportunities side by side, a trend line for applications over time and a KPI card for the Q4 2025 versus Q1 2026 comparison.

2.4 Design Choices

- Bar charts were used for comparing opportunities and categories, since they make it easy to see which one is highest or lowest.
- Time series chart was used for both the 6-month trend and Last 7-dates trend, showing direction and change better than a bar chart does.
- KPI cards were used for headline numbers to give at-a-glance summary.
- Every chart has a clear title and axis labels, so it can be understood without reading this report first.
- Also, Blank opportunity names were filtered out, and the 'Group in Others' option was turned off, so the chart shows only real, named opportunities sorted by sign-up count.

Total Application	Total Opportunity	Opportunity Category	Completion rate
14,825	5,730	10	7.63%

3. Visualizations and Validation

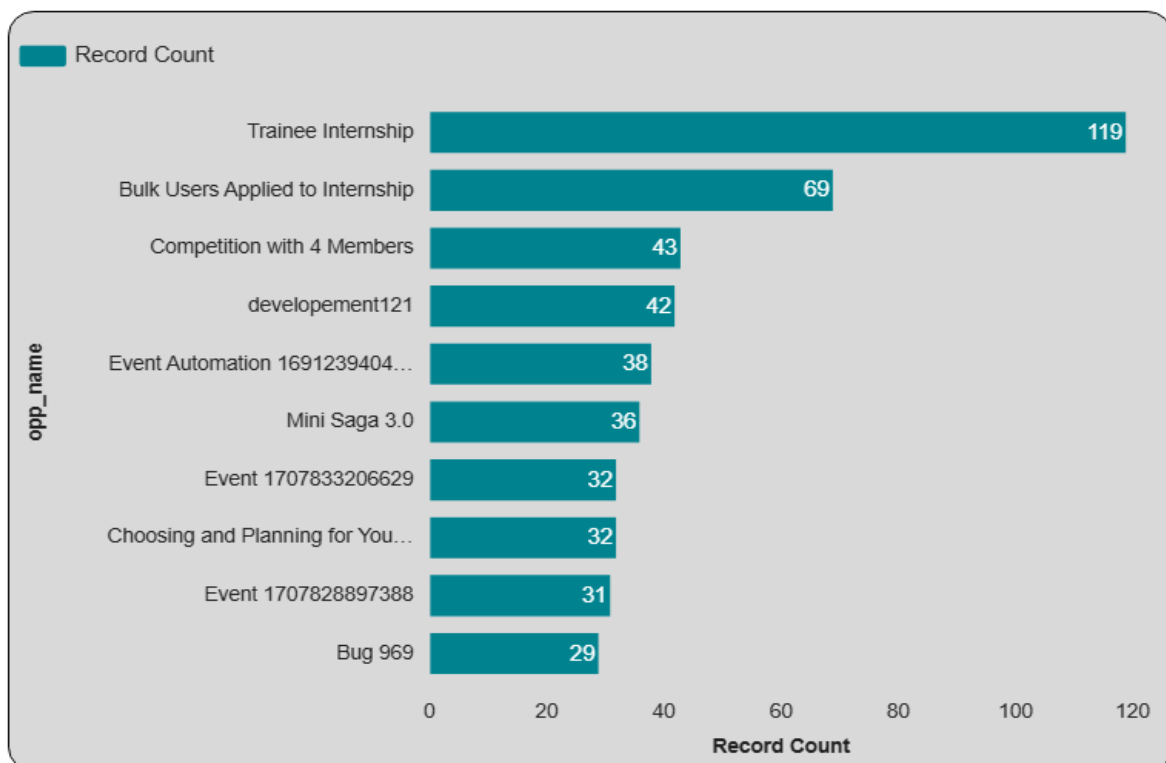
Each chart below shows what is on the dashboard, the SQL query used to check it, and the result of that query. The dashboard numbers matched the SQL results in every case.

3.1 Opportunity Sign-Up Performance

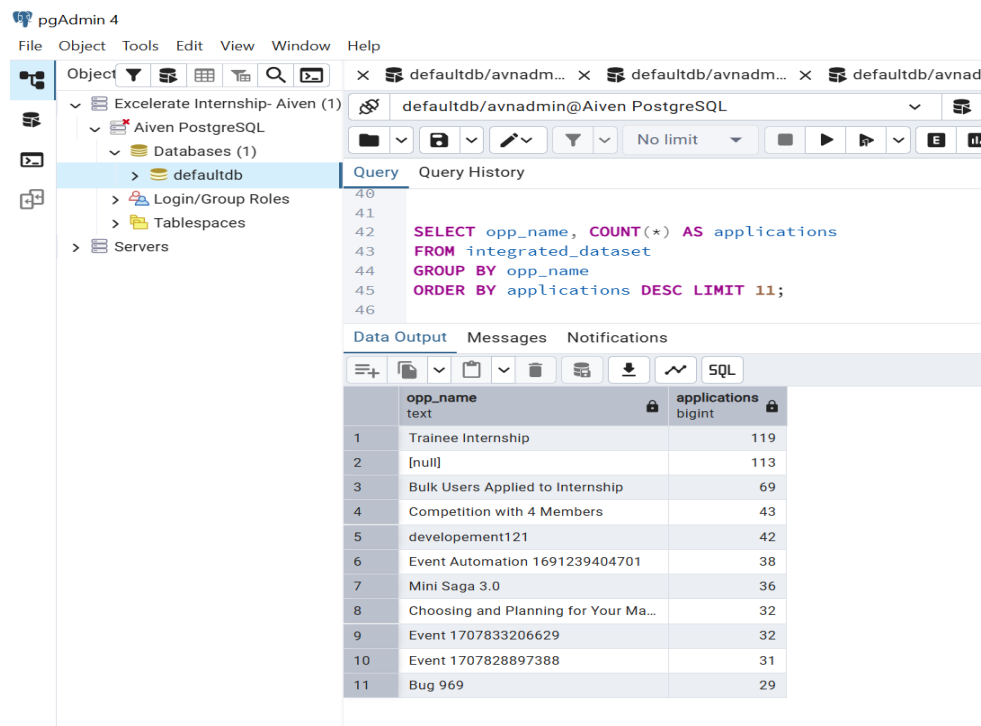
Chart type: Bar chart · Dimension: category · Metric: Record Count · Source: integrated_dataset

Purpose: Sign-up performance shows which opportunities get the most attention. This matters because it points out where demand is concentrated, and it sets up the comparison with completion rate in the next chart.

Chart 1: Opportunities by sign-up



```
SELECT opp_name, COUNT(*) AS applications
FROM integrated_dataset
GROUP BY opp_name
ORDER BY applications DESC
LIMIT 10;
```



Top 10 opportunities by sign-ups.

Trainee Internship has the most sign-ups by far, with 119 applications. Bulk Users Applied to Internship is next with 69. Most opportunities in the catalogue get only 1 or 2 applications. A small number of opportunities get most of the attention.

3.2 Completion Rates Across Opportunities

Chart type: Bar chart. Dimension: category. Metric: calculated field *Is complete*. Source: *integrated_dataset*

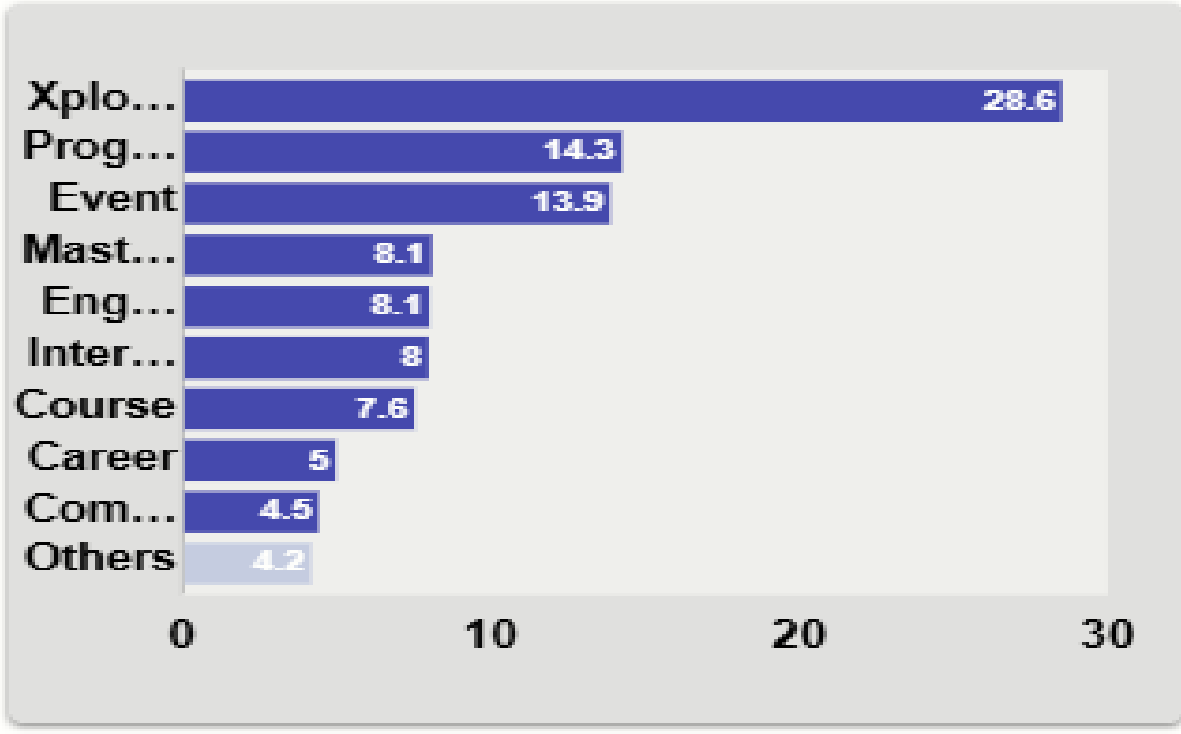
Purpose: shows what share of applications in each category actually reached completion, a measure of quality/follow-through, not just volume.

```

SELECT opp_name,
       COUNT(*) AS applications,
       COUNT(completion_date) AS completions,
       ROUND(100.0 * COUNT(completion_date) / COUNT(*), 1) AS completion_rate
FROM integrated_dataset
GROUP BY opp_name
HAVING COUNT(*) >= 5
ORDER BY completion_rate DESC LIMIT 10;

```

Complete Rate by category



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File Object Tools Edit View Window Help

Object x defaultdb/avnadm... x defaultdb/avnadm... x defaultdb/avnadm...

Excelerate Internship- Aiven (1)

Aiven PostgreSQL

Databases (1)

defaultdb

Login/Group Roles

Tablespaces

Servers

Query Query History

46 SELECT

47 category,

Data Output Messages Notifications

	category text	applications bigint	completions bigint	completion_rate numeric
1	Xploreu	7	2	28.6
2	Program	7	1	14.3
3	Event	1745	243	13.9
4	Masterclass	910	74	8.1
5	Engagement	942	75	8.0
6	Internship	5160	415	8.0
7	Course	1319	100	7.6
8	[null]	72	5	6.9
9	Career	2049	100	4.9
10	Competition	2195	98	4.5
11	JobSimulati...	415	18	4.3
12	NULL	4	0	0.0

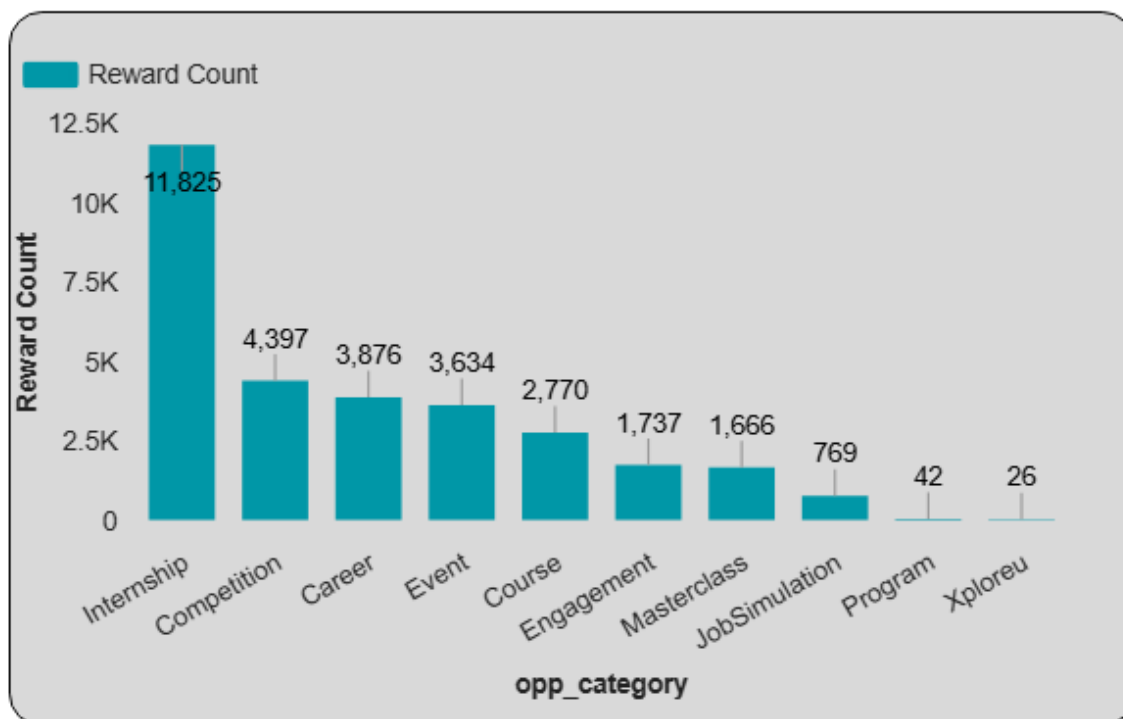
Events have the highest completion rate at 13.9%. Career and Competition have the lowest, both under 5%. The overall completion rate across every application is 7.6%. This is low, and it is worth looking into further in Week 4.

3.3 Reward Distribution

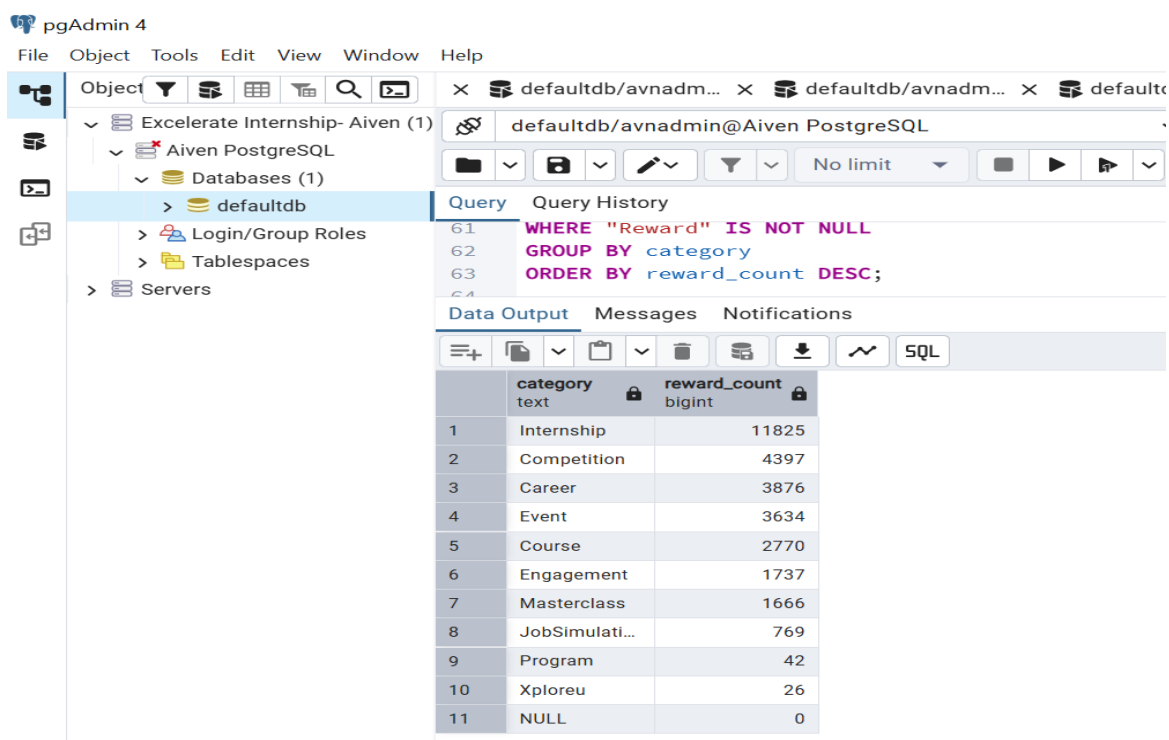
Chart type: Bar chart · Dimension: opportunity_category · Metric: Reward Count · Source: opportunity_data

Purpose: confirms how completely reward information is populated across the opportunity catalogue.

Chart 3: Reward Distribution by opp_category



```
SELECT opp_name, (LENGTH("Reward") - LENGTH(REPLACE("Reward", 'Reward#', '')))
  / LENGTH('Reward#') AS reward_count FROM integrated_dataset WHERE "Reward" IS
NOT NULL GROUP BY opp_name, "Reward"
ORDER BY reward_count DESC
LIMIT 10;
```



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File Object Tools Edit View Window Help

Object Explorer: Excelerate Internship- Aiven (1) > Aiven PostgreSQL > Databases (1) > defaultdb > Login/Group Roles > Tablespaces > Servers

Query: defaultdb/avnadmin@Aiven PostgreSQL

```

61 WHERE "Reward" IS NOT NULL
62 GROUP BY category
63 ORDER BY reward_count DESC;
64

```

Data Output Messages Notifications

	category text	reward_count bigint
1	Internship	11825
2	Competition	4397
3	Career	3876
4	Event	3634
5	Course	2770
6	Engagement	1737
7	Masterclass	1666
8	JobSimulati...	769
9	Program	42
10	Xploreu	26
11	NULL	0

Reward Distribution

The Reward field holds a block of text listing every reward given for an opportunity, not a plain number. It was counted by finding how many times the text “Reward#” appears. Across the whole catalog, there are 10,330 reward records. Internship has the most, at 2,947. This lines up with Internship also having the most applications overall.

3.4 Application Counts, Last 7 Days

Chart type: Time series chart · Dimension: apply_date (Date) · Metric: Record Count ·

Purpose: show how application volume has moved in the last 7 days in the program's records.

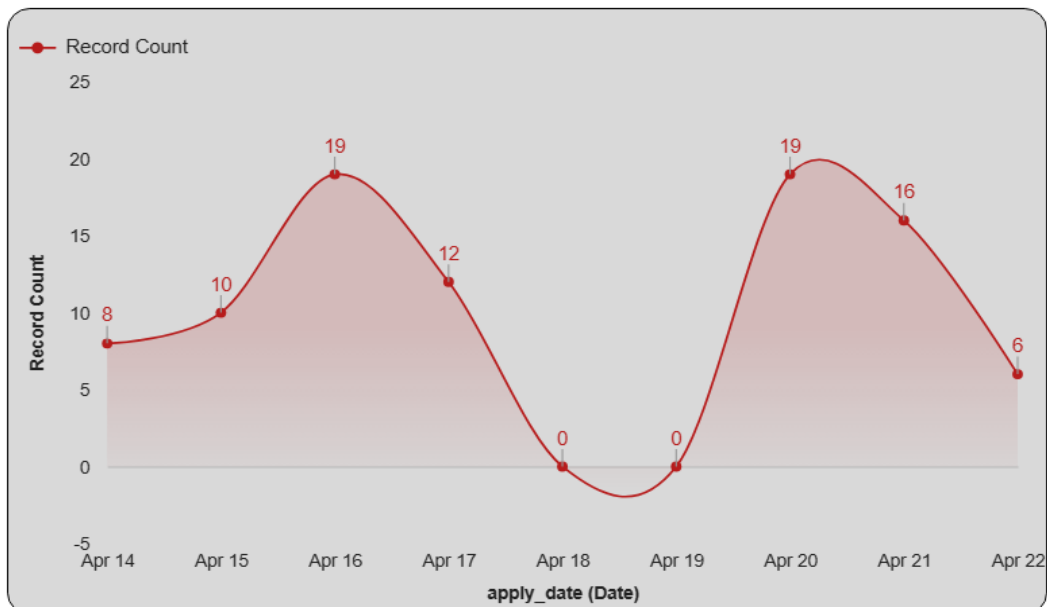
```

SELECT apply_date::date AS day, COUNT(*) AS applications
FROM integrated_dataset
WHERE apply_date IS NOT NULL
GROUP BY day
ORDER BY day DESC
LIMIT 7;

```

Daily applications during this period range from 6 to 19, with no clear pattern across the seven days.

Chart 4: Application Trend, Last 7 Dates



pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer: defaultdb/avnadm... x defaultdb/avnadm... x defaultdb/

Excelerate Internship- Aiven (1)

Aiven PostgreSQL

Databases (1)

defaultdb

Login/Group Roles

Tablespaces

Servers

Query: defaultdb/avnadmin@Aiven PostgreSQL

Query History

```

12 FROM integrated_dataset
13 WHERE apply_date IS NOT NULL
14 GROUP BY day
15 ORDER BY day DESC
16 LIMIT 7;
    
```

Data Output Messages Notifications

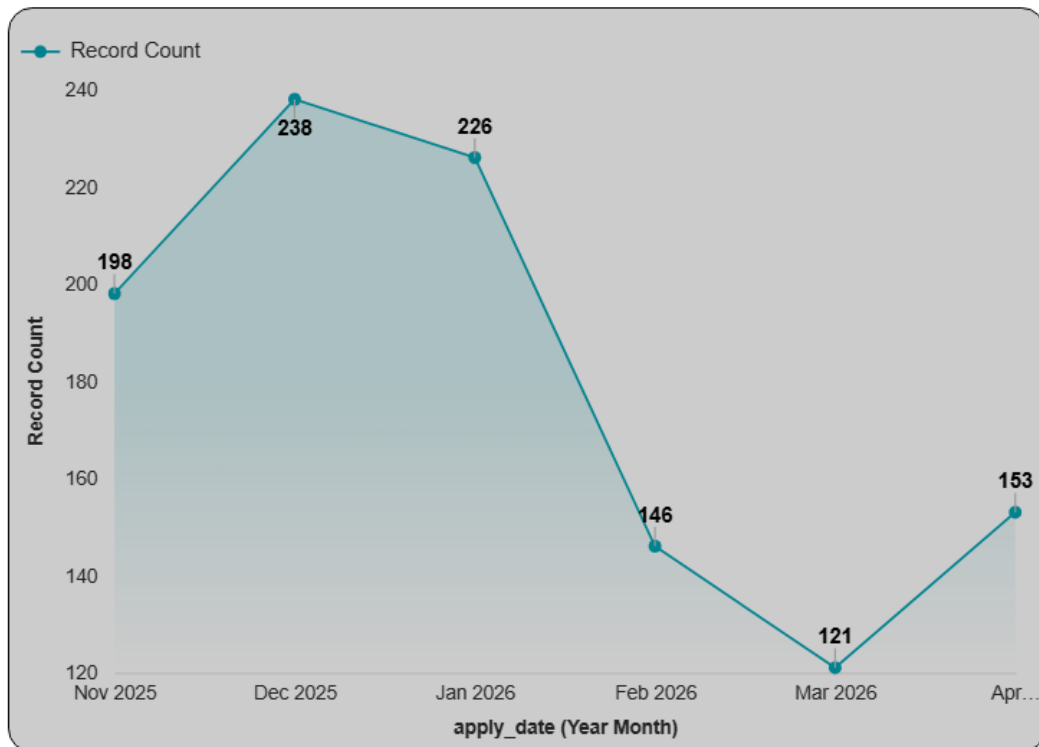
	day date	applications bigint
1	2026-04-22	6
2	2026-04-21	16
3	2026-04-20	19
4	2026-04-17	12
5	2026-04-16	19
6	2026-04-15	10
7	2026-04-14	8

3.5 Application Trends, Last 6 Months

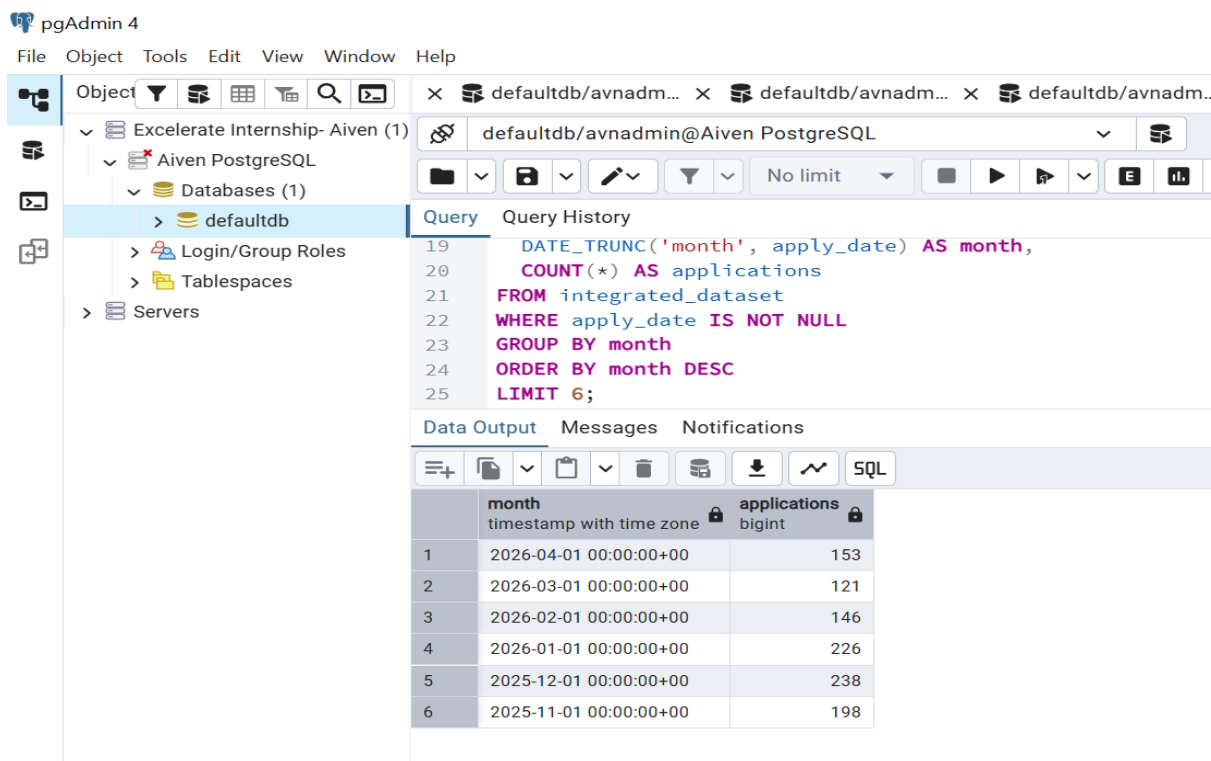
Chart type: Time series chart· Dimension: apply_date (Year, Month)· Metric: Record Count·

Purpose: show how application volume has moved month after month since the program's earliest records.

Chart 5: Application Trend, Last 6 Months



```
SELECT
    DATE_TRUNC('month', apply_date) AS month,
    COUNT(*) AS applications
FROM integrated_dataset
WHERE apply_date IS NOT NULL
GROUP BY month
ORDER BY month DESC
LIMIT 6;
```



Application trend for the last 6 months

Applications peaked in December 2025 at 238, then dropped each month after that. By March 2026 they were down to 121, before ticking back up slightly to 153 in April.

3.6 Scholarship Distribution by Category

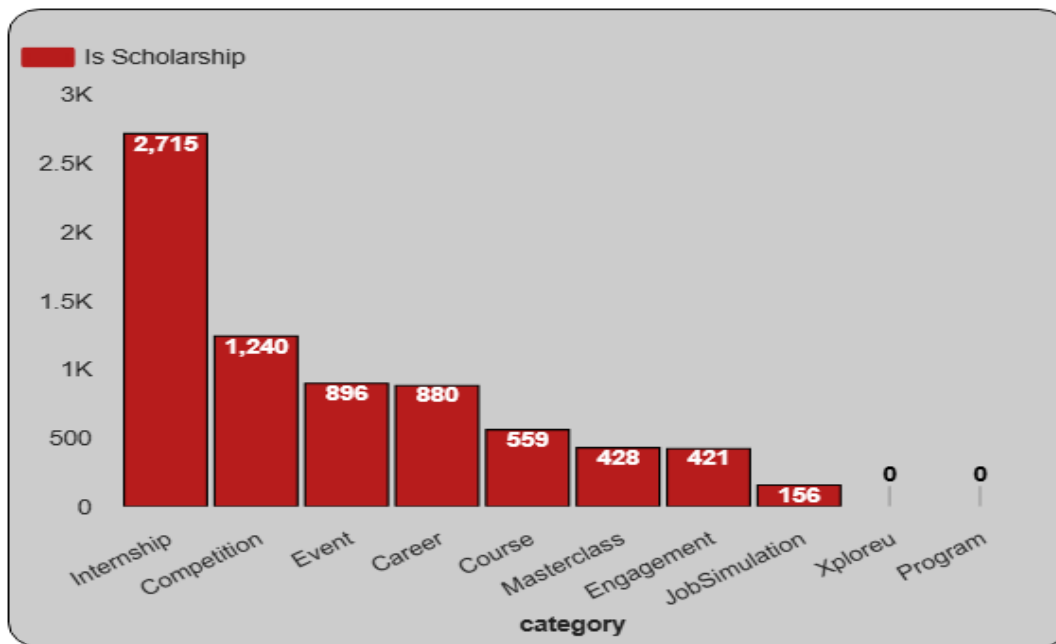
Chart type: Bar chart · Dimension: category · Metric: Sum of calculated field Is scholarship, sorted descending.

Purpose: show how many people in each category actually got a scholarship, like which category is handing out the most scholarships overall. scholarship-backed opportunities exist within each category.

```

SELECT category, COUNT(*) FILTER (WHERE amount_to_be_paid::numeric = 0) AS
scholarship_count
FROM integrated_dataset GROUP BY category ORDER BY scholarship_count DESC;

```

Chart 6: Scholarship Distribution by category

pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer: defaultdb/avnadm... x defaultdb/avnadm... x defaultdb/

- Excelerate Internship- Aiven (1)
 - Aiven PostgreSQL
 - Databases (1)
 - defaultdb
 - Login/Group Roles
 - Tablespaces
 - Servers

Query Editor: defaultdb/avnadmin@Aiven PostgreSQL

Query:

```

83 COUNT(*) FILTER (WHERE amount_to_be_paid::numeric
84 FROM integrated_dataset
85 GROUP BY category
86 ORDER BY scholarship_count DESC;

```

Data Output:

	category text	scholarship_count bigint
1	Internship	2715
2	Competition	1240
3	Event	896
4	Career	880
5	Course	559
6	Masterclass	428
7	Engagement	421
8	JobSimulati...	156
9	[null]	35
10	Program	0
11	NULL	0
12	Xploreu	0

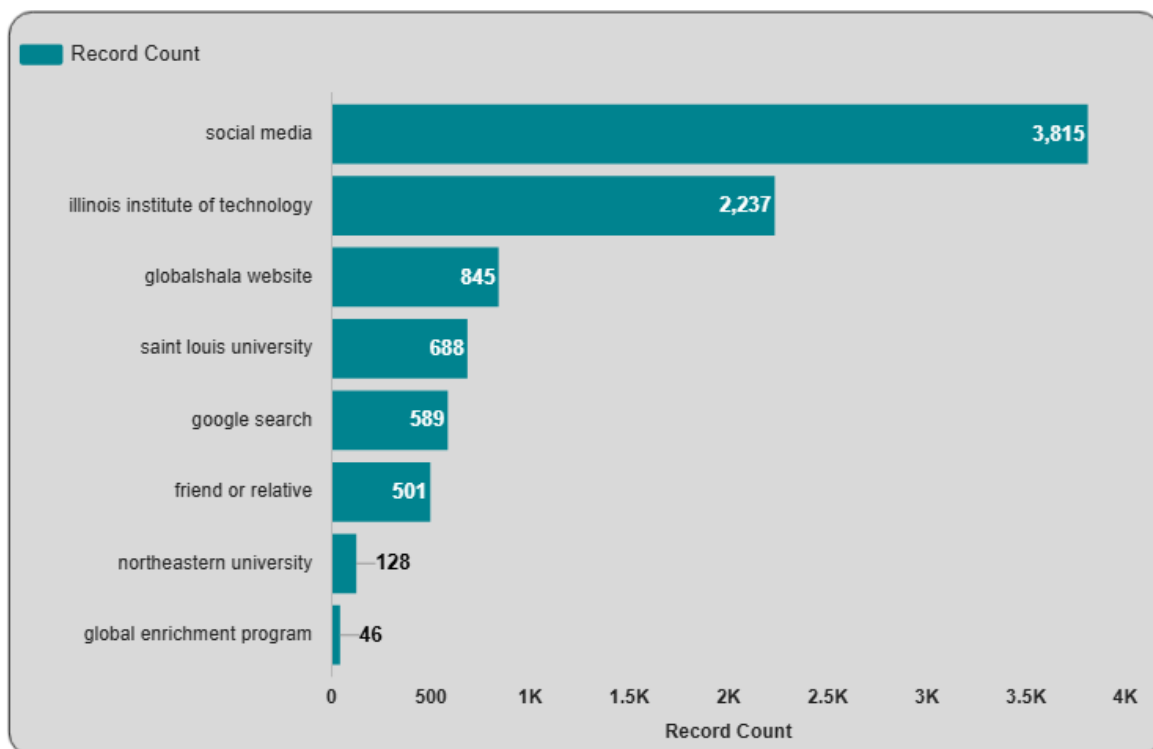
Competition has the highest scholarship distribution of 2,175. JobSimulation has the lowest at 156.

3.7 Lead Source Performance

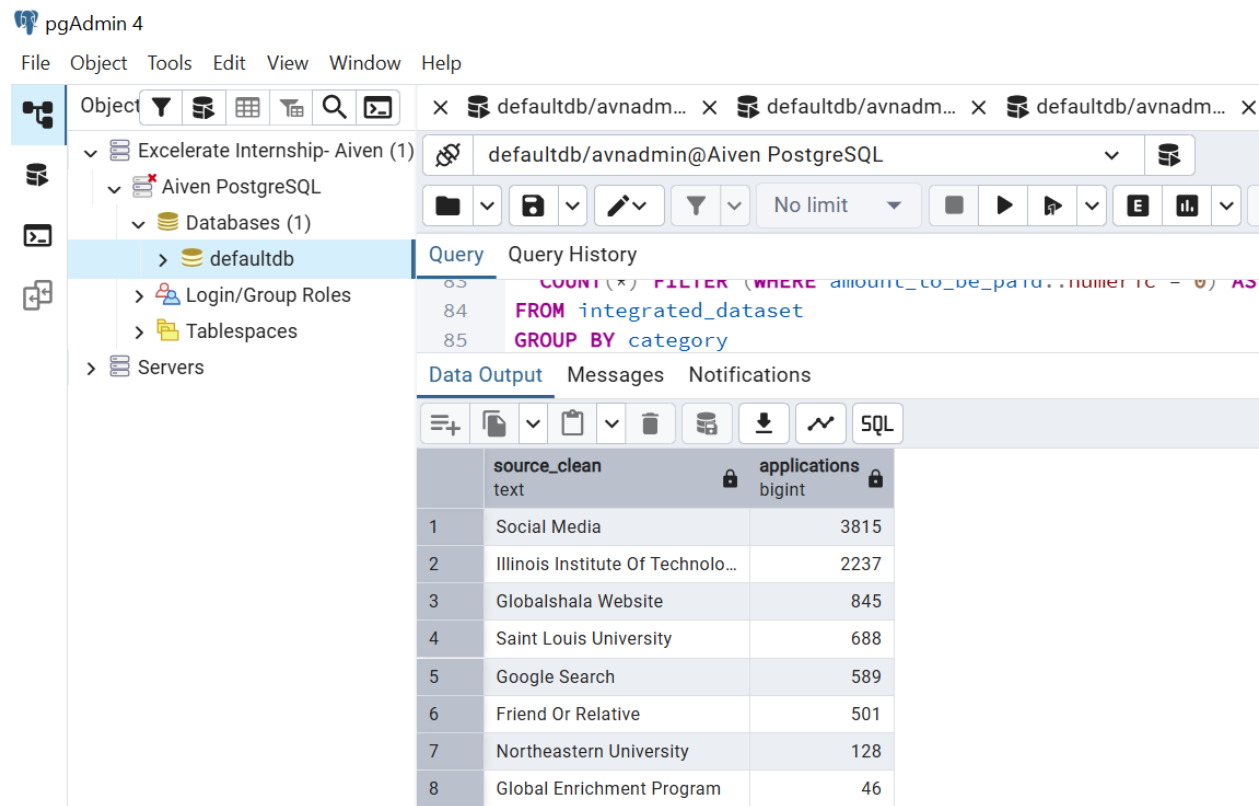
Chart type: Bar chart. Dimension: from_where_did_you_hear_about_u · Metric: Record Count, sorted descending.

Purpose: identifies which acquisition/referral channels are driving the most applications. Before charting, near-duplicate labels caused by inconsistent capitalization and spelling (e.g. "social media" vs "Social Media"; "my friend or relative" vs "my friend/relative"; "other" vs "others") were standardized in SQL so the chart reflects true totals per channel.

Chart 7: Lead Source Performance



```
SELECT from_where_did_you_hear_about_us, COUNT(*) AS applications
FROM integrated_dataset
WHERE from_where_did_you_hear_about_us IS NOT NULL
GROUP BY from_where_did_you_hear_about_us
ORDER BY applications DESC
LIMIT 10;
```



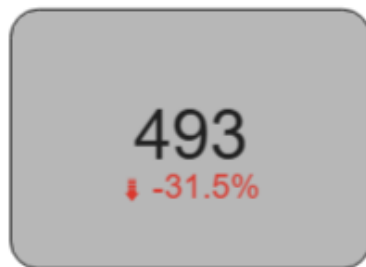
Application by lead source

Social Media is the biggest source by far, at 3,815 applications, which is 42.8% of all applications with a known source. Illinois Institute of Technology is next at 2,237 applications, or 25.1%. Together, these two sources make up more than two thirds of all applications.

3.8 2025 Q4 vs 2026 Q1 Comparison

Chart 8: 2025 Q4 vs 2026 Q1 Comparison

2026 Q1 Application



```

SELECT
  COUNT(*) FILTER (
    WHERE apply_date >= '2025-10-01' AND apply_date < '2026-01-01'
  ) AS q4_2025,
  COUNT(*) FILTER (
    WHERE apply_date >= '2026-01-01' AND apply_date < '2026-04-01'
  ) AS q1_2026
FROM integrated_dataset;

```

2025 Q4 had 720 applications. 2026 Q1 had 493. That is a drop of 31.5%. This matches the monthly trend in Section 3.5, which also shows a steady decline starting in December 2025.

Percentage change = $(493 - 720) / 720 \times 100 = -31.5\%$

pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer: Excelsite Internship- Aiven (1) > Aiven PostgreSQL > Databases (1) > defaultdb

Query Editor: defaultdb/avnadmin@Aiven PostgreSQL

Query:

```

SELECT
  COUNT(*) FILTER (WHERE apply_date >= '2025-10-01' A
  COUNT(*) FILTER (WHERE apply_date >= '2026-01-01' A
FROM integrated_dataset;

```

Data Output:

	q4_2025 bigint	q1_2026 bigint
1	720	493

4. Insights

4.1 Which opportunities perform best?

Trainee Internship has the most applications by far, but its completion rate is not captured in the completion list, which means a lot of people apply but fewer finish. Smaller opportunities like Event 1707833206629 and Event 1707828897388 have very high completion rates, above 96%, but far fewer applications. This suggests that smaller, more focused opportunities may keep learners more engaged than large, high-volume ones.

4.2 Where are engagement levels strongest or weakest?

By category, Event has the strongest completion rate at 13.9%. Career and Competition are the weakest, both under 5%. The overall completion rate across all applications is 7.6%, which is low. This means most applications never reach a completed state, regardless of category.

4.3 Which lead sources contribute the most activity?

Social Media is the dominant acquisition channel (3,815 applications) with a known source, more than any other source. Illinois Institute of Technology is second at 2,237. Together these two sources account for more than two out of every three applications. The remaining sources, including Google Search and word of mouth, each contribute a much smaller share.

4.4 Are there notable changes over time?

Yes. Applications peaked in December 2025 at 238, then fell every month after that through March 2026, which dropped to 121. April 2026 showed a small recovery to 153, but it is still well below the December peak. Comparing full quarters confirms this: 2025 Q4 had 720 applications, and 2026 Q1 had only 493, a drop of 31.5%.

5. Recommendations

- Investigate why Career and Competition have such low completion rates compared to Event. Interviewing a few learners who did not finish could show whether the issue is time friction, too much workload, clash of interest, or something else.
- Investigate the drop in applications since December 2025. Check if this lines up with a seasonal pattern, a marketing change, or something on the platform side. The Q4 to Q1 drop of 31.5% is large enough to be worth a closer look.
- Since Social Media and Illinois Institute of Technology bring in most applications, consider whether other partner institutions could be reached the same way. Relying on two sources for most activity is a risk if either one slows down.
- Consider whether high-volume opportunities like Trainee Internship need a different follow-up process than smaller ones, since their completion rates appear lower. A check-in step partway through the opportunity might help more learners finish.
- Continue tracking the categories with lower completion in future weeks, to see whether any changes made actually improve the rate.

Conclusion

Week 3 turned the cleaned, integrated dataset from Week 2 into a working dashboard. The dashboard covers all six areas requested in the brief: sign-up performance, completion rates, reward distribution, application trends, scholarship distribution, and lead source performance. Every number on the dashboard was checked against a matching SQL query, and all of them matched.

The data shows a few clear patterns. A small number of opportunities get most of the sign-ups. Completion rates are low overall, and vary a lot by category. Two lead sources account for most applications. Applications have dropped since December 2025. These findings are turned into specific recommendations in Section 5.